Metals & Ceramics

This course focuses on 4 main classes of materials:

Metals & Alloys

Pros: High stiffness, strength, ductility, toughness & thermal and electrical conductivity
Cons: High density & poor corrosion resistance (unless other alloying processes employed)

Used for load bearing structures, frames, engines, etc.

Ceramics

Pros: High hardness, stiffness, compressive strength, thermal stability & corrosion resistance
Cons: brittle and low toughness

Used in furnace lining, tiles, cutting tools, break pads, etc.

Polymers & Elastomers

Pros: Low density, friction, electrically insulating and corrosion resistant
Cons: Low stiffness, strength

Used in consumer packaging, healthcare devices, etc.

Composites

Pros: (generally) high strength to weight ratio and corrosion resistance, but properties are tailored to use case
Cons: expensive, difficult to repair and recycle.

Used in aircraft parts, boat hulls, etc.

Material properties are due to the composition, structure and bonding of the constituent elements.

Metals

metal atoms undergo metallic bonding in the material. metallic bonding consists of a regular structure of metal ions in a sea of delocalised electrons.

Delocalised electrons give rise to thermal and electrical conductivity, when a thermal or voltage gradient is applied. However, delocalised electrons can be easily removed, causing the poor corrosion resistance (unless alloying or other processes are applied)

Metal ions also pack them in a regular crystalline structure, and since metal ions tend to be heavier, the added dense crystalline structure causes the high density of metals.

The delocalised sea of electrons also allows layers of metal ions to slide past one another under shear forces, allowing for the metal to be deformed, giving high ductility and malleability.

Ceramics

Ceramics are either nonmetallic-nonmetallic compounds or nonmetallic-metallic compounds. More specifically, ceramics are either ionic or covalent compounds.

Ionic bonding is due to the transfer of valence electrons, causing two ions to electrostatically attract one another to form an ionic bond. The ionic bonds are non-directional, and pack themselves in a regular crystalline structure.

Covalent bonding is due to the sharing of valence electrons, thus being a directional bond. This causes specific and highly ordered crystal structures.

Since ceramics use non-metallic elements, the low atomic number of these causes the lower density of the ceramic. Especially for covalent ceramics, the packing is not close packed, adding on to the low density.

The electrons for both ionic and covalent ceramics are held with strong ionic/covalent bonds respectively, making it hard to be removed by the environment.

Unlike metals, when a shear force is applied, the ions slide past another until like ions are brought closer to each other, causing electrostatic repulsion between the like ions, causing brittleness. Likewise for covalent ceramics, the directional bonds resist changes to the configuration of the atoms, also causing brittleness.

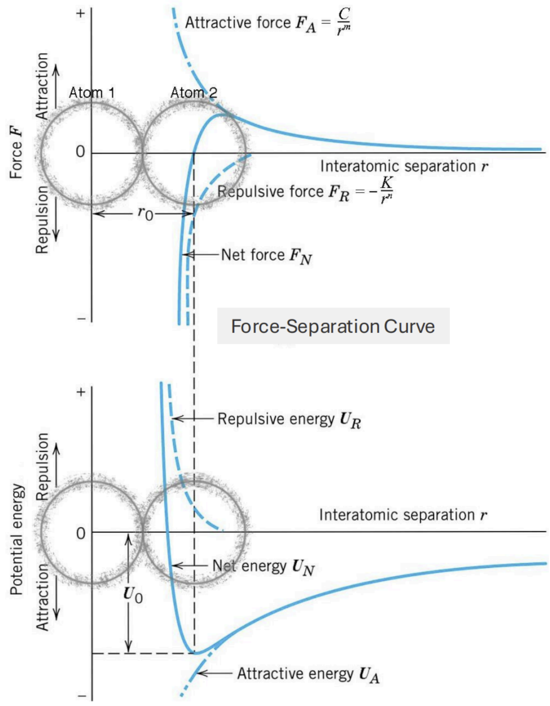
Since a high amount of energy is needed to shear the ceramic, but a low amount of strain produced, ceramics have low toughness.

The strong ionic and covalent bonds as well as the low plastic deformation of the ceramic causes the high hardness of ceramics.

Bonding Energies and Forces

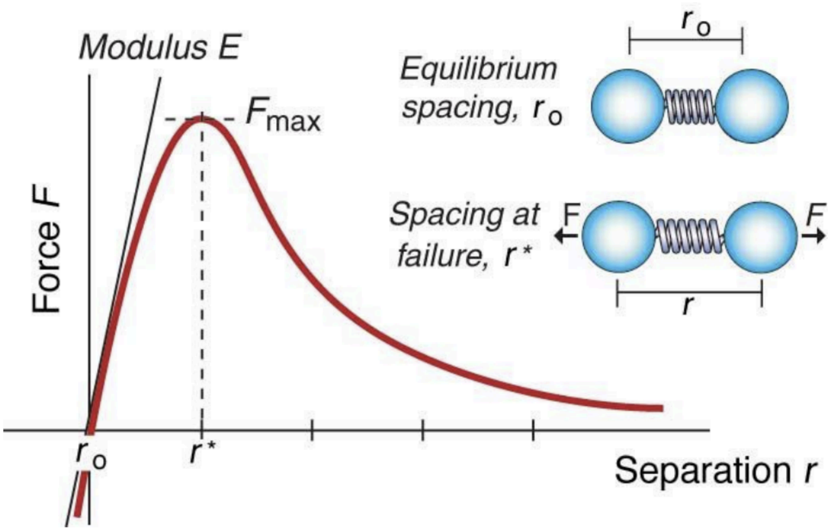
Atoms bond with one another as it reduces their thermodynamic energy, making it more stable. As such, when atoms bond, a certain amount of energy, called the bond energy, is released.

If two atoms are pulled apart, there is an interatomic force, $F = \frac{dU}{dr}$, that restores equilibrium. We can model this interatomic force as a spring between two atoms, that exerts a force to pull or push the atoms back to equilibrium. When no external force is applied, the atoms are at equilibrium and the bonding energy is at a minimum.



Bonding energies can explain other mechanical properties.

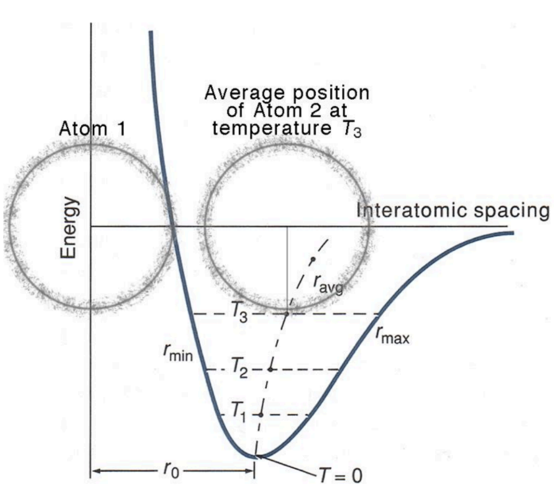
For example, say a material is extended, which causes the atoms to separate. The bonds' separation is then determined by the force-separation curve.

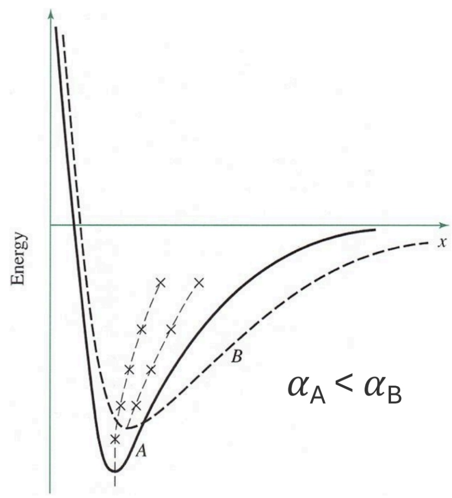


For separations where r has not exceeded the failure spacing r*, the material deforms elastically, hence at the start, the initial slope of the force separation curve is indicative of the Young’s Modulus. However, once the separation exceeds the spacing at failure, the bonds break, which refers to the theoretical strength of the material.

The bonds can also be used to explain thermal expansion, as the bonds cause the atoms to vibrate about their positions under some heating, where their amplitude increases with heating.

As this amplitude is defined by the energy-separation curve, more specifically the well of the curve, which is asymmetric, as temperature increases, the average position they vibrate about increases, causing expansion.





Furthermore, for stiffer bonds between the atoms, the steeper the energy-separation curve, and thus the more narrower and deeper the well, causing lesser thermal expansion.

Similarly, a shallower well also means that with enough heating, the amplitude can be high enough to escape the well, causing the bonds to break. Hence, a higher melting point just means the bonds are stiffer and the energy-separation has a deeper well, requiring more energy to escape the well.